

The health department wants to identify where high uninsurance overlaps with poor health outcomes and what realistic actions can be taken given limited resources. My approach was staged: descriptive ranking to benchmark performance, overlap analysis to find counties facing multiple problems at once, and regression to identify the strongest socioeconomic predictors. This framework kept the analysis consistent from beginning to end.

Starting with descriptive ranking in Excel, Los Alamos, New Mexico emerged as a consistent top performer, ranking among the best in both uninsured rate and poor health percentage. Lincoln, South Dakota appeared in the top 5 for poor health, physically unhealthy days, and mentally unhealthy days. On the other end, Texas counties dominated the worst uninsured rates and worst poor health rates, while West Virginia stood out in mentally unhealthy days.

To identify overlapping challenges, I took the 10 counties with the highest uninsured rates and mapped how they ranked across other health outcomes using a heatmap. Kenedy, Texas stood out most severely, ranking #1 in uninsured rate, #1 in poor health, and #2 in physically unhealthy days — the only county with three outcomes in the worst 100 nationally. Presidio and Hidalgo, Texas each had two outcomes in the worst 100, both ranking in the top 10 worst for uninsured rate and poor health.

Using multiple linear regression in Python across three models — physically unhealthy days, mentally unhealthy days, and percent fair or poor health — education-related variables were consistently the strongest predictors. Percent Some College had the strongest negative relationship in the physical and mental models, meaning counties with higher college attainment tended to have fewer unhealthy days. This also helps explain why Los Alamos and Lincoln, both higher-attainment counties, consistently performed well across measures.

Kenedy, Presidio, and Hidalgo should be considered the highest-priority counties for intervention. All three rank in the bottom 7 percent nationally for unemployment, which matters because 63 percent of working-age adults receive insurance through employers (Kolb et al., 2025). High unemployment likely contributes to high uninsured rates, reducing access to care and contributing to worse outcomes. Since the health department cannot directly address unemployment, the most practical short-term actions are expanding insurance enrollment outreach and increasing access to care through mobile clinics or community health workers in these counties. Long term, since education was the strongest predictor across all three models, the health department should support broader cross-sector investment in college attainment and educational opportunity, even though those structural issues extend beyond the department's direct control.

AI (ChatGPT and Claude) was used for debugging, refining visualization code, and improving wording. Specifically, it helped generate and refine matplotlib code for heatmap and regression visualizations. All data collection, analysis design, interpretation, and conclusions are our own.

References

Kolb, K., Radley, D. C., & Collins, S. R. (2025, March 13). *How affordable is job-based health coverage for workers?* The Commonwealth Fund.

<https://www.commonwealthfund.org/blog/2025/how-affordable-job-based-health-coverage-workers>